

Azure Fundamentals Lab Manual

Hands-on Lab: Virtual Machine, Azure SQL Database, Blob Storage, and a Simple HTML Website

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All exercises use the Azure Portal (no command line required).

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Introduction

This lab manual is a guided, hands-on introduction to four core Azure services. Every step is performed in the Azure Portal using point-and-click navigation — no scripting or command-line tools are required. By the end of this lab you will have created a virtual machine, an Azure SQL database, a storage account with image upload, and a simple HTML website hosted on Azure.

What You Will Build

Lab	Outcome
Lab 1: Virtual Machine	A running Windows or Linux VM you can connect to remotely
Lab 2: Azure SQL Database	A cloud SQL database with a server, accessible via Query editor
Lab 3: Blob Storage	A storage account and container with an image file uploaded
Lab 4: Simple HTML Deployment	A static HTML page published to a public URL

Prerequisites

- An active Azure subscription (a free trial or pay-as-you-go subscription both work).
- Permission to create resources in at least one resource group.
- A modern web browser (Edge, Chrome, or Firefox).
- One small image file (JPG or PNG) saved on your computer, for Lab 3.
- A short HTML file, for Lab 4 (a sample is provided in Lab 4).

Warning — Cost Control

Every resource created in this lab manual incurs cost while it is running. Follow the Cleanup Checklist at the end of this document to delete resources when you are finished, especially the virtual machine and the SQL database.

Conventions Used in This Manual

- Numbered steps describe actions to take in the Azure Portal.
- Yellow Note boxes provide helpful context or alternatives.
- Red Warning boxes flag cost or security considerations.
- Text in quotation marks (e.g. “Create a resource”) refers to exact buttons, menu items, or labels in the portal.

Lab 1: Create a Virtual Machine

In this lab, you will deploy a virtual machine (VM) running Windows Server or Ubuntu Linux directly from the Azure Portal.

Step 1: Sign In and Start the Wizard

1. Go to the Azure Portal at portal.azure.com and sign in with your Azure account.
2. In the search bar at the top of the page, type “Virtual machines” and select it from the results.
3. On the Virtual Machines page, select “+ Create”, then choose “Azure virtual machine” from the dropdown.

Step 2: Configure the Basics Tab

4. Under Project details, select your Subscription.
5. For Resource group, select “Create new” and enter a name, for example LabResourceGroup.
6. Under Instance details, enter a Virtual machine name, for example MyLabVM.
7. Choose a Region close to you.
8. For Image, select a Windows Server image (e.g. Windows Server 2022 Datacenter) or Ubuntu Server (e.g. Ubuntu Server 22.04 LTS).
9. Leave the default Size, or select “See all sizes” and choose a small, low-cost size such as B1s for lab purposes.
10. Under Administrator account, enter a Username and a Password (or upload an SSH public key if using Linux). The password must be at least 12 characters and meet complexity requirements.
11. Under Inbound port rules, allow the relevant inbound port: RDP (3389) for Windows, or SSH (22) for Linux.

[Basics](#)
[Disks](#)
[Networking](#)
[Management](#)
[Monitoring](#)
[Advanced](#)
[Tags](#)
[Review + create](#)

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Training-2026

Resource group * ⓘ

(New) RG_Dharshini_K

[Create new](#)

Instance details

Virtual machine name * ⓘ

MyLabVM

Region * ⓘ

(Asia Pacific) South India

[Deploy to an Azure Extended Zone](#)

Availability options ⓘ

No infrastructure redundancy required

Security type ⓘ

Trusted launch virtual machines

[Configure security features](#)

Image * ⓘ

 Ubuntu Server 24.04 LTS - x64 Gen2

[See all images](#) | [Configure VM generation](#)

VM architecture ⓘ

☐ Arm64
 ☒ x64

< Previous

Next : Disks >

Review + create

 [Give feedback](#)

Step 3: Review and Create

12. Select “Review + create” at the bottom of the page.
13. Wait for validation to pass, then review the configuration summary.
14. Select “Create”.
15. Wait for the deployment to complete (typically 1–3 minutes), then select “Go to resource”.

Basics

Subscription	Training-2026
Resource group	(new) RG_Dharshini_K
Virtual machine name	MyLabVM
Region	South India
Availability options	No infrastructure redundancy required
Zone options	Self-selected zone
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No
Image	Ubuntu Server 24.04 LTS - Gen2
VM architecture	x64
Size	Standard B2als v2 (2 vcpus, 4 GiB memory)
Enable Hibernation	No
Authentication type	SSH public key
Username	azureuser
SSH Key format	RSA
Key pair name	MyLabVM_key
Public inbound ports	SSH
Azure Spot	No

Your deployment is complete



Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-2026...

Subscription : [Training-2026](#)

Resource group : [RG_Dharshini_K](#)

Start time : 30/6/2026, 10:35:12 AM

Correlation ID : 765d8f7f-45dd-4974-91db-c0cd624cddfb

> Deployment details

✓ Next steps

[Set up auto-shutdown](#) Recommended

[Monitor VM health, performance, and network dependencies](#) Recommended

[Run a script inside the virtual machine](#) Recommended

[Go to resource](#)

[Create another VM](#)

[Scale out your VM](#)

Step 4: Connect to the VM

16. On the VM Overview page, select "Connect".
17. For Windows, choose RDP, download the RDP file, and open it with Remote Desktop Connection using the username and password you set.
18. For Linux, choose SSH and follow the connection command shown, using an SSH client such as PuTTY or a terminal.

Note

If RDP or SSH connection fails, confirm the inbound port rule was created and that your network allows outbound connections on that port.

```
warning: Permanently added '20.219.74.250' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1018-azure x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

```
System information as of Tue Jun 30 05:12:11 UTC 2026
```

```
System load:  0.0      Processes:            130
Usage of /:    5.8% of 28.02GB Users logged in:      0
Memory usage:  7%      IPv4 address for eth0: 10.3.0.4
Swap usage:    0%
```

Connected to linux via VM.

Lab 2: Create an Azure SQL Database

In this lab, you will create a logical SQL server and a single Azure SQL Database, then query it using the built-in Query editor — entirely in the browser.

Step 1: Start the Wizard

19. In the Azure Portal search bar, type “SQL databases” and select it.
20. Select “+ Create”.

Step 2: Configure the Basics Tab

21. Select your Subscription.
22. For Resource group, select the same resource group used in Lab 1, or create a new one.
23. Under Database details, enter a Database name, for example mySampleDatabase.
24. Next to Server, select “Create new”. Enter a unique Server name, choose a Location, select an authentication method (SQL authentication is simplest for a lab), and set a Server admin login and Password. Select “OK”.
25. Leave “Want to use SQL elastic pool?” set to No.
26. Under Compute + storage, select “Configure database”, choose the General Purpose tier with Serverless compute, and select the smallest vCore option to minimize cost. Select “Apply”.
27. Under Backup storage redundancy, choose Locally-redundant backup storage (lowest cost for a lab).

Step 3: Configure Networking

28. Select the “Networking” tab.

29. For Connectivity method, select “Public endpoint”. **(automatically done in newer version)**
30. Set “Allow Azure services and resources to access this server” to Yes.
31. Set “Add current client IP address” to Yes so you can connect from your own machine.

Basics

Subscription	Training-2026
Resource group	RG_Dharshini_K
Region	centralindia
Database name	sampledb
Server	dharshini-server-central
Compute + storage	General Purpose - Serverless: Standard-series (Gen5), 1 vCore, 16 GB storage, zone redundant disabled
Backup storage redundancy	Locally-redundant backup storage

Networking

Allow Azure services and resources to access this server	Yes
Private endpoint	None

Security

Transparent data encryption (Server level)	Service-managed key selected
Database level customer-managed key	Not configured
Database level user assigned managed identity	Not configured
Advanced data security	Not now
Always Encrypted with secure enclaves	Not configured
Sql Ledger(Database)	Disabled
Digest Storage	Disabled

Step 4: Review, Create, and Query

32. Select “Review + create”, wait for validation, then select “Create”.

Your deployment is complete

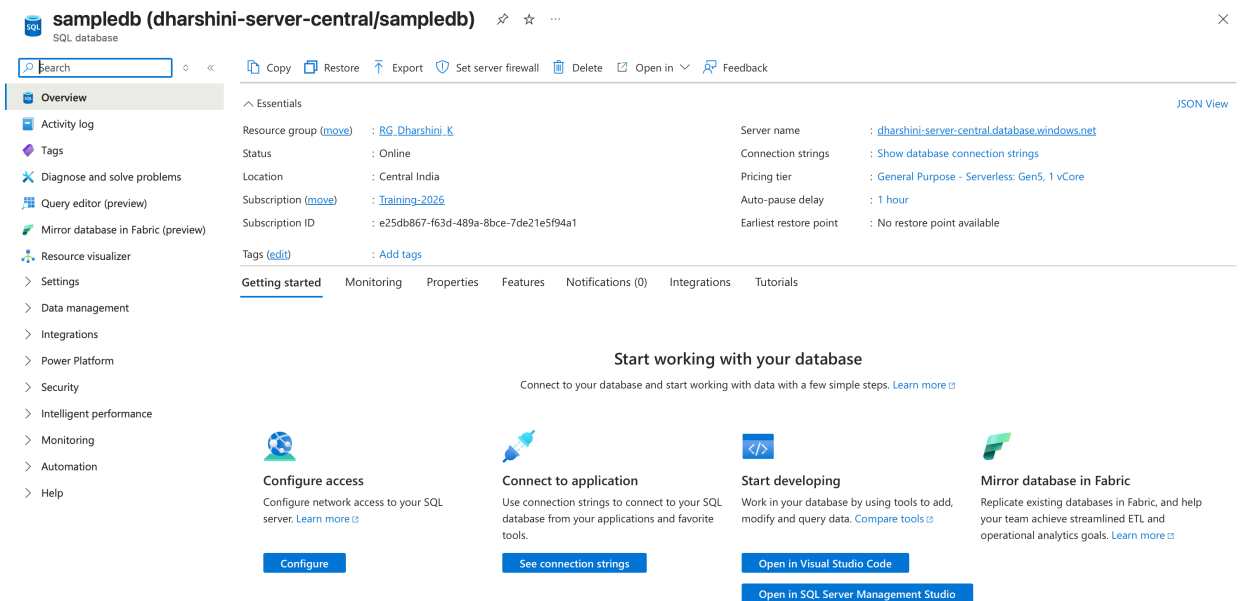
 Deployment name : Microsoft.SQLDatabase.newDatabaseExistingServer... Start time : 30/6/2026, 11:14:41 AM
 Subscription : [Training-2026](#) Correlation ID : d222c14e-d009-44ac-9a15-9040e3cff36b
 Resource group : [RG_Dharshini_K](#)

[Deployment details](#)

[Next steps](#)

[Go to resource](#)

33. Once deployment finishes, select “Go to resource”.



sampledb (dharshini-server-central/sampledb) SQL database

Search

Copy Restore Export Set server firewall Delete Open in Feedback

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Query editor (preview)
- Mirror database in Fabric (preview)
- Resource visualizer
- Settings
- Data management
- Integrations
- Power Platform
- Security
- Intelligent performance
- Monitoring
- Automation
- Help

Essentials

Resource group (move) : [RG_Dharshini_K](#)

Status : Online

Location : Central India

Subscription (move) : [Training-2026](#)

Subscription ID : e25db867-f63d-489a-8bce-7de21e5f94a1

Tags (edit) : [Add tags](#)

Server name : [dharshini-server-central.database.windows.net](#)

Connection strings : [Show database connection strings](#)

Pricing tier : General Purpose - Serverless: Gen5, 1 vCore

Auto-pause delay : 1 hour

Earliest restore point : No restore point available

Tags (edit) : [Add tags](#)

Getting started Monitoring Properties Features Notifications (0) Integrations Tutorials

Start working with your database

Connect to your database and start working with data with a few simple steps. [Learn more](#)

Configure access
Configure network access to your SQL server. [Learn more](#)

Connect to application
Use connection strings to connect to your SQL database from your applications and favorite tools.

Start developing
Work in your database by using tools to add, modify and query data. [Compare tools](#)

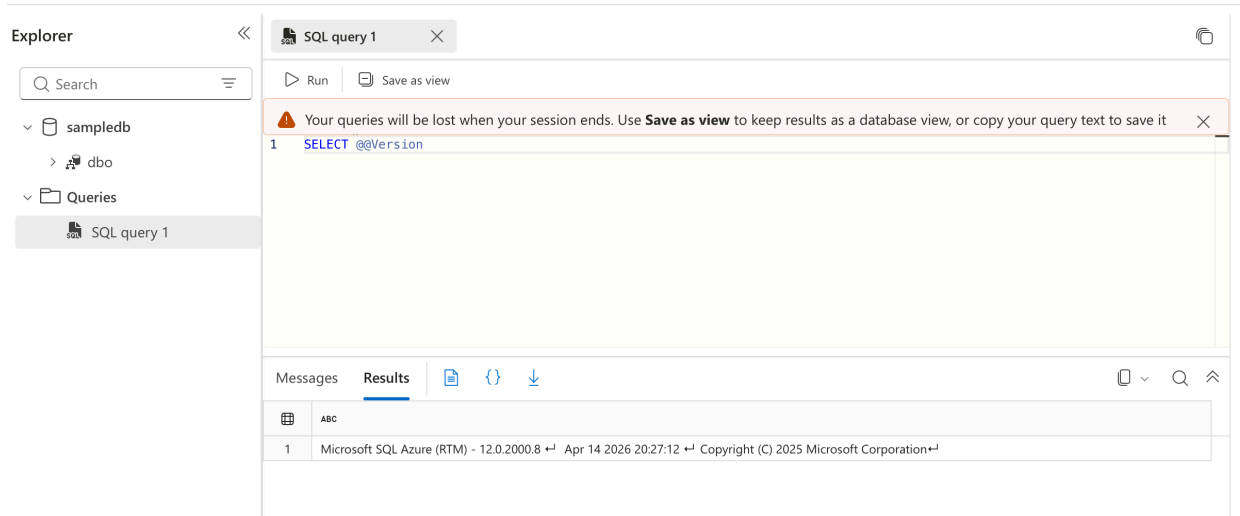
Mirror database in Fabric
Replicate existing databases in Fabric, and help your team achieve streamlined ETL and operational analytics goals. [Learn more](#)

[Configure](#) [See connection strings](#) [Open in Visual Studio Code](#) [Open in SQL Server Management Studio](#)

34. On the database Overview page, select “Query editor (preview)” in the left menu.

35. Sign in using the SQL admin login and password you created.

36. In the query window, run a simple test query, for example: `SELECT @@VERSION;`, then select “Run” to confirm the database is working.

**Note**

The Query editor runs entirely in the browser, so no database client software is required for this lab.

Lab 3: Blob Storage and Image Upload

In this lab, you will create a storage account, add a container, and upload an image file as a blob.

Step 1: Create a Storage Account

37. In the Azure Portal search bar, type “Storage accounts” and select it.
38. Select “+ Create”.
39. Select your Subscription and the resource group from Lab 1.
40. Enter a globally unique Storage account name (lowercase letters and numbers only, e.g. mylabstorage2026).
41. Choose a Region close to you.
42. Leave Performance set to Standard and Redundancy set to Locally-redundant storage (LRS) for a low-cost lab setup.

Create a storage account ...

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below.

[Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription *	Training-2026
Resource group *	RG_Dharshini_K Create new

Instance details

Storage account name * ⓘ	labstoragedharshini
Region * ⓘ	(Asia Pacific) South India Deploy to an Azure Extended Zone
Preferred storage type	Choose preferred storage type
i This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more	
Performance * ⓘ	☒ Standard: Recommended for most scenarios (general-purpose v2 account) ☐ Premium: Recommended for scenarios that require low latency.
Redundancy * ⓘ	Locally redundant storage (LRS)

43. Select “Review”, then “Create”, and wait for deployment to finish.

Create a storage account ...

Basics

Subscription	Training-2026
Resource group	RG_Dharshini_K
Location	South India
Storage account name	labstoragedharshini
Primary service	
Performance	Standard
Replication	Locally redundant storage (LRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed Identity for SMB	Disabled

Azure Files

Require Encryption in Transit for SMB	Enabled
---------------------------------------	---------

Security

Secure transfer	Enabled
Blob anonymous access	Disabled
Allow storage account key access	Enabled
Default to Microsoft Entra authorization in	Disabled

[Previous](#)[Next](#)[Create](#)

44. Select "Go to resource".

✓ Your deployment is complete



Deployment name : labstorageharshini_1782799149884

Subscription : [Training-2026](#)

Resource group : [RG_Dharshini_K](#)

Start time : 30/6/2026, 11:29:44 AM

Correlation ID : a918e39e-49d8-489f-9f13-8e4162eb7c67

> Deployment details

✓ Next steps

[Go to resource](#)

labstorageharshini Storage account

Search [] Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser
- Storage Mover
- Partner solutions
- Resource visualizer
- Data storage
- Security + networking
- Data management
- Settings
- Monitoring
- Automation
- Help

Essentials

Resource group (move) : [RG_Dharshini_K](#)

Location : southindia

Subscription (move) : [Training-2026](#)

Subscription ID : e25db867-f63d-489a-8bce-7de21e5f94a1

Disk state : Available

Tags (edit) : [Add tags](#)

Performance : Standard

Replication : Locally redundant storage (LRS)

Account kind : StorageV2 (general purpose v2)

Provisioning state : Succeeded

Created : 30/6/2026, 11:29:45 AM

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

- Hierarchical namespace : Disabled
- Default access tier : Hot
- Blob anonymous access : Disabled
- Blob soft delete : Enabled (7 days)
- Container soft delete : Enabled (7 days)
- Versioning : Disabled
- Change feed : Disabled
- NFS v3 : Disabled
- Allow cross-tenant replication : Disabled
- Storage tasks assignments : None

File service

- Identity-based access : Not configured
- Default hierarchical namespace : Disabled

Security

- Require secure transfer for REST API operations : Enabled
- Storage account key access : Enabled
- Minimum TLS version : Version 1.2
- Infrastructure encryption : Disabled

Networking

- Public network access : Enabled
- Public network access scope : Enable from all networks
- Private endpoint connections : 0
- Network routing : Microsoft network routing
- Endpoint type : Standard

Add or remove favorites by pressing Cmd+SHIFT+F

Step 2: Create a Container

45. In the left menu of the storage account, under Data storage, select “Containers”.
46. Select “+ Container”.
47. Enter a name, for example images (lowercase letters, numbers, and dashes only).
48. Leave the default Public access level (Private), then select “Create”.

Showing all 2 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector	
☐	\$logs	30/6/2026, 11:30:14 AM	Private	Available	-	...
☐	images	30/6/2026, 11:33:02 AM	Private	Available	-	...

Step 3: Upload an Image

49. Select the new container to open it.
50. Select “Upload”.
51. Browse to the image file on your computer (JPG or PNG) and select it.
52. Select “Upload” to complete the operation.
53. Select the uploaded blob in the list to view its properties and copy its URL.

pet1.png

Blob

Save Discard Download Refresh Delete Change tier Acquire lease Break lease Give feedback

Overview Versions Snapshots Edit Generate SAS

Properties

URL	https://labstoragedharsh...
LAST MODIFIED	30/6/2026, 11:45:33 AM
CREATION TIME	30/6/2026, 11:45:33 AM
VERSION ID	-
TYPE	Block blob
SIZE	148.16 KiB
ACCESS TIER	Hot (Inferred)
ACCESS TIER LAST MODIFIED	N/A

Copied

Note

If the container's public access level is Private, the blob URL will not be publicly viewable in a browser. To make an individual image viewable via its URL, set the container access level to “Blob” (anonymous read access for blobs only) when creating it, or change it afterward under container Settings > Access level.

Lab 4: Deploy a Simple HTML Website

In this lab, you will publish a basic HTML page to the public internet using the static website feature of the storage account you created in Lab 3 — all from the portal, with no command-line deployment tools.

Step 1: Prepare a Sample HTML File

Create a text file named `index.html` on your computer with the following content, then save it:

```
<!DOCTYPE html> <html> <head><title>My Azure Lab Site</title></head> <body>  <h1>Hello from  
Azure Blob Storage!</h1>  <p>This page was deployed using static website hosting.</p> </body>  
</html>
```

Step 2: Enable Static Website Hosting

54. Open the storage account created in Lab 3.
55. In the left menu, under Data management, select “Static website”.
56. Select “Enabled”.
57. For Index document name, enter `index.html`.
58. Select “Save”.
59. Azure displays a Primary endpoint URL once saved — copy this URL; it is the public address of your site.

 Save  Discard |  Give feedback

Enabling static websites on the blob service allows you to host static content. Webpages may include static content asynchronously from primary to secondary regions, files at the secondary endpoint may not be immediately available.

Static website

Disabled **Enabled**

An Azure Storage container has been created to host your static website.

[\\$web](#)

 Improve the page load time of your static website by using the caching features of Azure Front Door (Advanced).

Primary endpoint 

<https://labstorageedharshini.z30.web.core.windows.net/>

Index document name 

index.html

Error document path 

Step 3: Upload the HTML File

60. After saving, return to the Static website page and select the link next to “\$web container” (Azure automatically creates a container named \$web for static content).

61. Select “Upload”.

62. Browse to your index.html file and select it.

63. Select “Upload” to complete the operation.

 \$web

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

 Add filter

 Search blobs by prefix (case-sensitive) Only show active blobs

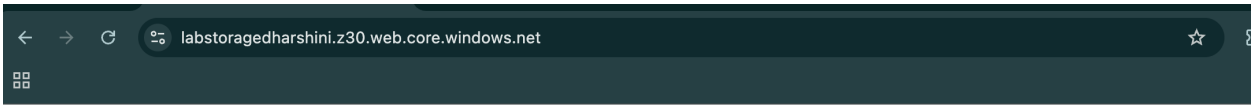
Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	 index.html	30/6/2026, 11:55:33 AM	Hot (Inferred)	Block blob	430 B	Available 

Step 4: View the Site

64. Return to the Static website page in the left menu of the storage account.

65. Select or copy the Primary endpoint URL and open it in a new browser tab.
66. Confirm that your HTML page renders correctly.



This Document is hosted in Azure Cloud through Blob Storage

This image is redired from azure blob storage container



Note

The \$web container is automatically configured for public, anonymous read access when static website hosting is enabled, so your page is reachable by anyone with the URL.

Cleanup Checklist

Azure resources continue to incur charges until they are deleted. When you have finished the lab, clean up resources to avoid unnecessary cost.

Option A: Delete the Entire Resource Group (Recommended)

Resource Group:

Resources

Recent Favorite

Name	Type	Last Viewed
 labstorage-dharshini	Storage account	13 minutes ago
 MyLabVM	Virtual machine	37 minutes ago
 RG_Dharshini_K	Resource group	37 minutes ago
 sampled-b (dharshini-server-central/sampled-b)	SQL database	41 minutes ago
 dharshini-server-central	SQL server	54 minutes ago
 mylabvm943	Network interface	an hour ago
 MyLabVM-nsg	Network security group	an hour ago
 MyLabVM-ip	Public IP address	an hour ago
 MyLabVM-vnet	Virtual network	an hour ago

[See all](#)

67. In the Azure Portal search bar, type “Resource groups” and select it.
68. Select the resource group used throughout this lab (e.g. LabResourceGroup).
69. Select “Delete resource group”.
70. Type the resource group name to confirm, then select “Delete”.

Delete a resource group



The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

 RG_Dharshini_K 

6 resources being deleted across 1 subscriptions, 1 resource groups

All dependent resources, including hidden types, are shown

Name	Resource type	Subscription	Resource Group
 dharshini-server-central	SQL server	Training-2026	RG_Dharshini_K
 MyLabVM	Virtual machine	Training-2026	RG_Dharshini_K
 MyLabVM-ip	Public IP address	Training-2026	RG_Dharshini_K
 MyLabVM-nsg	Network security group	Training-2026	RG_Dharshini_K
 MyLabVM-vnet	Virtual network	Training-2026	RG_Dharshini_K
 mylabvm943	Network interface	Training-2026	RG_Dharshini_K

Enter resource group name to confirm deletion *

Delete

Cancel

Deleting the resource group removes the VM, SQL server and database, storage account, and all associated resources (disks, network interfaces, public IP addresses) in one step.

Option B: Delete Resources Individually

- Virtual machine: open the VM, select “Delete”, and also delete the associated disk, network interface, public IP, and network security group shown on the VM's Overview page.
- SQL Database and server: open the SQL server resource, select “Delete” (this also offers to delete databases on that server).
- Storage account: open the storage account, select “Delete”, and confirm by typing the account name.

Warning — Cost Control

A virtual machine continues to bill for compute time while running, and for attached disks even when stopped. Always delete or fully deallocate resources you no longer need.

Summary

In this lab manual you created and configured four foundational Azure resources entirely through the Azure Portal: a virtual machine, an Azure SQL Database, a Blob Storage container with an uploaded image, and a static HTML website. These exercises cover compute, database, and storage — three of the core pillars of the Azure platform — and provide a foundation for more advanced labs involving networking, security, and automation.

Further Reading

[Create a Windows VM in the Azure portal](#)

[Create a single Azure SQL database \(portal\)](#)

[Upload, download, and list blobs \(Azure portal\)](#)

[Host a static website in Azure Storage](#)